Com S 4720/5720 Fall 2025

Problem Set 6 (35 pts)

Due at 11:59pm

Friday, October 10

All the problems are from *https://aimacode.github.io/aima-exercises/* (possibly) with minor typo fixes, some rewriting, or additions of missing information.

1.(10 pts) (Exercise 6.5) Solve the cryptarithmetic problem in Figure 6.2 below by hand, using the strategy of backtracking with forward checking and the MRV and least-constraining-value heuristics.

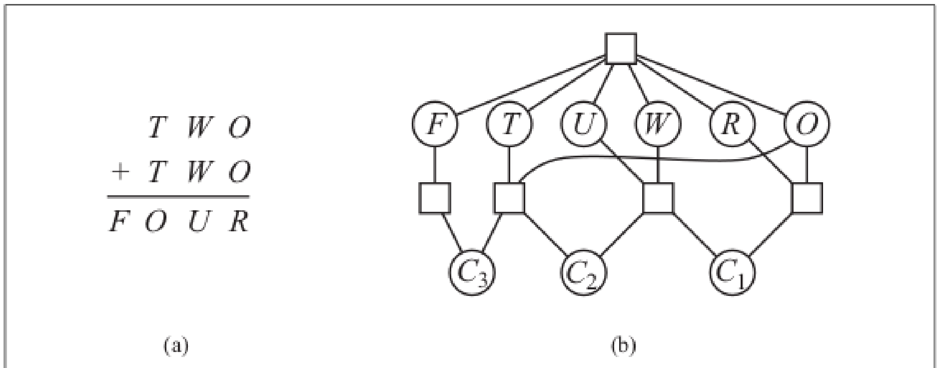

Figure 6.2 (a) A cryptarithmetic problem. Each letter stands for a distinct digit; the aim is to find a substitution of digits for letters such that the resulting sum is arithmetically correct, with the added restriction that no leading zeroes are allowed. (b) The constraint hypergraph for the cryptarithmetic problem, showing the *Alldiff* constraint (square box at the top) as well as the column addition constraints (four square boxes in the middle). The variables C_1, C_2, and C_3 represent the carry digits for the three columns.

2.(16 pts)(Exercise 7) Consider the following logic puzzle. In five houses, each with a different color, live five persons of different nationalities, each of whom prefers a different brand of candy, a different drink, and a different pet. Given the following fourteen facts, the questions to answer are "Where does the zebra live, and in which house do they drink water?"

1. The Englishman lives in the red house.

2. The Spaniard owns the dog.

3. The Norwegian lives in the first house on the left.

4. The green house is immediately to the right of the ivory house.
5. The man who eats Hershey bars lives in the house next to the man with the fox.
6. Kit Kats are eaten in the yellow house.
7. The Norwegian lives next to the blue house.
8. The Smarties eater owns snails.
9. The Snickers eater drinks orange juice.
10. The Ukrainian drinks tea.
11. The Japanese eats Milky Ways.
12. Kit Kats are eaten in a house next to the house where the horse is kept.
13. Coffee is drunk in the green house.
14. Milk is drunk in the middle house.

We can formulate the problem as a constraint satisfaction problem. The five houses are numbered 1, 2, 3, 4, and 5 from left to right. These are all the values to be assigned in our formulation.

(a) (8 pts) How many variables are there? List all of them.

(b) (8 pts) Rewrite the fourteen facts as constraints in terms of these variables.

3.(9 pts) (Exercise 6.11) Use the AC-3 algorithm to show that arc consistency can detect the inconsistency of the partial assignment $\{WA = green, V = red\}$ for the map-coloring problem shown in Figure 6.1 on the previous page.